

TIVAT, Montenegro

WMO: 134570

Lat: 42.405N

Lon: 18.723E

Elev: 20

StdP: 14.69

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 28.0	(c) 30.1	(d) 7.4	(e) 8.0	(f) 39.1	(g) 11.9	(h) 10.1	(i) 41.1	(j) 24.8	(k) 45.1	(l) 21.5	(m) 46.2	(n) 2.0	(o) 80	(p) 0.393

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 23.8	(c) 94.1	(d) 72.4	(e) 91.7	(f) 72.4	(g) 89.6	(h) 71.8	(i) 76.1	(j) 88.4	(k) 74.8	(l) 87.5	(m) 73.4	(n) 86.2	(o) 5.5	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
71.9	118.2	83.7	70.3	111.6	82.1	69.1	107.1	80.8	39.5	88.6	38.2	87.9	37.0	86.2	85.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 18.5	(b) 15.3	(c) 12.6	(e) 23.4	(f) 98.1	(g) 2.4	(h) 2.3	(i) 21.7	(j) 99.7	(k) 20.3	(l) 101.1	(m) 18.9	(n) 102.4	(o) 17.1	(p) 104.0		
			DB													
			WB													
				21.7	79.1	2.4	1.8	20.0	80.4	18.6	81.4	17.3	82.4	15.5	83.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	60.1	45.3	46.7	50.7	56.6	64.3	71.8	76.5	76.5	68.8	61.5	54.1	47.6
	DBStd	12.08	5.20	5.24	4.84	4.96	4.75	5.21	3.81	3.98	4.61	4.60	5.70	5.44
	HDD50	471	162	117	50	6	0	0	0	0	0	0	24	112
	HDD65	2911	611	513	444	256	72	8	0	0	16	124	327	540
	CDD50	4161	15	24	71	203	444	655	821	821	564	357	149	37
	CDD65	1125	0	0	0	3	50	213	356	356	130	16	1	0
	CDH74	13127	0	1	1	50	566	2524	4427	4256	1180	121	1	0
	CDH80	5464	0	0	0	6	92	948	2076	2041	297	4	0	0
Wind	WSAvg	3.8	4.4	4.9	5.0	3.9	3.4	3.2	3.5	3.2	3.3	3.2	3.8	4.5
Precipitation	PrecAvg	82.3	9.3	8.3	7.7	7.6	4.4	3.0	1.9	2.9	6.3	8.9	11.3	10.7
	PrecMax	119.3	20.8	16.6	18.5	13.8	8.3	9.7	9.4	10.0	15.6	19.8	21.5	16.6
	PrecMin	46.8	0.4	0.0	0.1	2.4	0.6	0.4	0.3	0.6	0.9	0.4	3.1	2.6
	PrecStd	17.6	5.3	3.9	3.7	3.0	2.1	2.2	2.3	2.6	3.7	4.3	3.8	4.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.3	65.9	70.0	79.6	86.0	94.6	96.9	97.2	89.6	79.8	71.9	64.2
		MCWB	53.4	53.7	54.9	61.8	67.9	73.3	73.1	72.1	69.7	66.5	61.2	55.8
	2%	DB	59.1	62.2	66.4	74.5	82.5	91.2	93.6	94.7	86.1	77.1	69.3	61.8
		MCWB	53.1	52.5	53.8	59.2	66.4	72.8	72.6	71.9	69.7	65.3	61.3	55.8
	5%	DB	57.1	59.3	64.2	71.3	79.9	88.1	91.6	91.9	83.6	75.0	67.2	59.7
		MCWB	51.8	51.3	53.7	57.8	65.5	71.6	72.5	72.1	68.8	64.0	60.1	54.1
	10%	DB	55.1	57.2	61.4	68.2	77.1	85.8	89.6	89.6	80.7	72.2	64.6	57.5
		MCWB	49.3	50.3	52.3	56.9	64.5	70.6	71.8	71.9	67.8	63.0	58.0	52.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.7	56.9	58.7	63.5	70.6	77.0	77.9	78.1	74.6	69.9	65.4	60.0
		MCDB	59.4	60.5	64.3	74.5	80.7	88.2	90.4	88.9	83.6	75.2	68.9	62.2
	2%	WB	54.9	54.9	57.0	61.5	68.4	74.9	76.1	76.1	72.3	68.0	63.2	57.6
		MCDB	57.8	59.1	62.8	71.3	79.2	87.4	88.9	88.3	81.6	73.6	67.5	60.5
	5%	WB	52.9	53.3	55.5	59.8	66.9	73.3	74.7	74.6	70.8	66.2	61.1	55.8
		MCDB	56.0	58.0	61.3	67.9	77.6	85.4	87.9	87.6	80.0	72.3	65.5	58.6
	10%	WB	51.0	51.7	54.2	58.4	65.4	71.7	73.3	73.2	69.2	64.6	59.2	53.9
		MCDB	53.8	56.3	59.8	65.7	75.1	83.7	86.6	86.4	78.3	70.7	63.6	56.9
Mean Daily Temperature Range		MDBR	17.1	18.0	18.8	19.8	21.3	22.5	23.8	24.1	21.5	20.5	18.2	16.9
	5% DB	MCDBR	19.4	21.6	23.6	24.6	25.5	25.9	27.2	27.8	24.8	24.0	20.6	18.9
		MCWBR	13.6	14.0	13.9	12.6	12.6	11.7	11.8	12.1	12.4	13.8	13.0	13.2
	5% WB	MCDBR	14.9	17.6	19.9	22.0	23.9	24.3	25.1	25.2	22.1	19.5	16.5	13.0
		MCWBR	11.8	12.8	13.4	12.6	13.0	12.2	12.7	13.1	12.7	12.7	11.5	10.5
Clear-Sky Solar Irradiance	taub		0.340	0.365	0.410	0.463	0.450	0.435	0.434	0.444	0.423	0.396	0.367	0.340
	taud		2.426	2.339	2.227	2.103	2.205	2.298	2.295	2.261	2.311	2.364	2.393	2.430
	Ebn at Noon		252	262	262	256	263	267	266	259	256	248	238	240
	Edh at Noon		25	32	40	49	45	41	41	41	36	31	26	23
All-Sky Solar Radiation	RadAvg		524	756	1157	1539	1913	2198	2265	2005	1436	980	606	475
	RadStd		59	119	161	162	149	138	121	122	103	95	62	85

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d) N/A	(e) N/A	(f) -3.47	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) -144	(l) +180	(m) N/A
Regional (22 neighbors)	+0.88	N/A	N/A	N/A	N/A	N/A	N/A	-191	+180	N/A

Nomenclature: See separate page